

Minesweeper

By Lexi Purser

A brief introduction to Minesweeper

Minesweeper is a puzzle game where players uncover a grid, avoiding hidden mines. Numbers on revealed squares show how many mines are nearby, helping players use logic to flag mines and clear safe spaces. The game is won when all mines have been flagged and the correct amount of flags are present.



Performance Measure

Penalties	Game End Rewards
• Click on invalid tile: -1 • Flag on invalid tile: -1 • Unflagging invalid tile: -1	Game won: $+\text{grid width}^2$
	Game lost: • Correct flag: +1 • Incorrect flag: -1

Environment

- **Observability:** Partially observable. Necessary starting information is available, however the location of the mines is not available.
- **Uncertainty:** Stochastic. Mine placement is random and unknown.
- **Duration:** Episodic. A past move is not relevant to us making our current move. We only know/need to know the current state of the game board.
- **Stability:** The game is static. Once you start a game, the mine positions remain the same until a new game is started.
- **Granularity:** Discreet.
- **Participants:** Single agent
- **Knowledge:** Unknown. Tile values are unknown (until we reveal them), however the rules and logic of the game is known

Actuators

- Choose game board difficulty
- Click on tile to reveal it, if it is a mine, you lose
- Flag a tile if you believe it hides a mine
- Unflag a tile if you have incorrectly placed a flag
- Use revealed numbers to know the number of adjacent mines
- Flag all mines (and nothing else) to win

Sensors

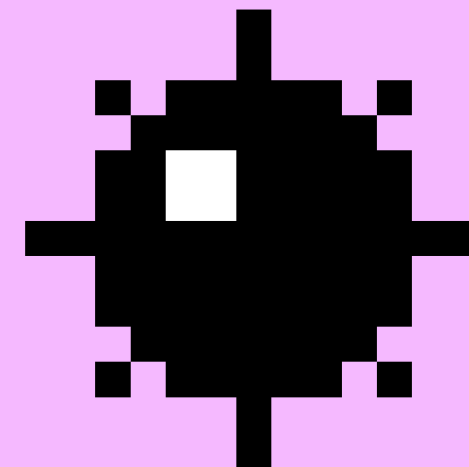


Tile status: cleared, flagged, or uncleared.

Value of revealed tile: This is an integer in the range $[0, 9]$.

Board size and mine count options:

- 9x9 board with 10 mines
- 16x16 board with 40 mines
- 22x22 board with 99 mines



Agent Design - No Guessing Mode

No guessing mode has been implemented in this agent.

On some versions of minesweeper there is a “No Guessing” mode. If you begin a game when this mode is active, it chooses your first action for you and automatically clears a tile that has zero mines adjacent to it. Any zero tiles adjacent to the selection will also be cleared by default.

Agent Design - Patterns

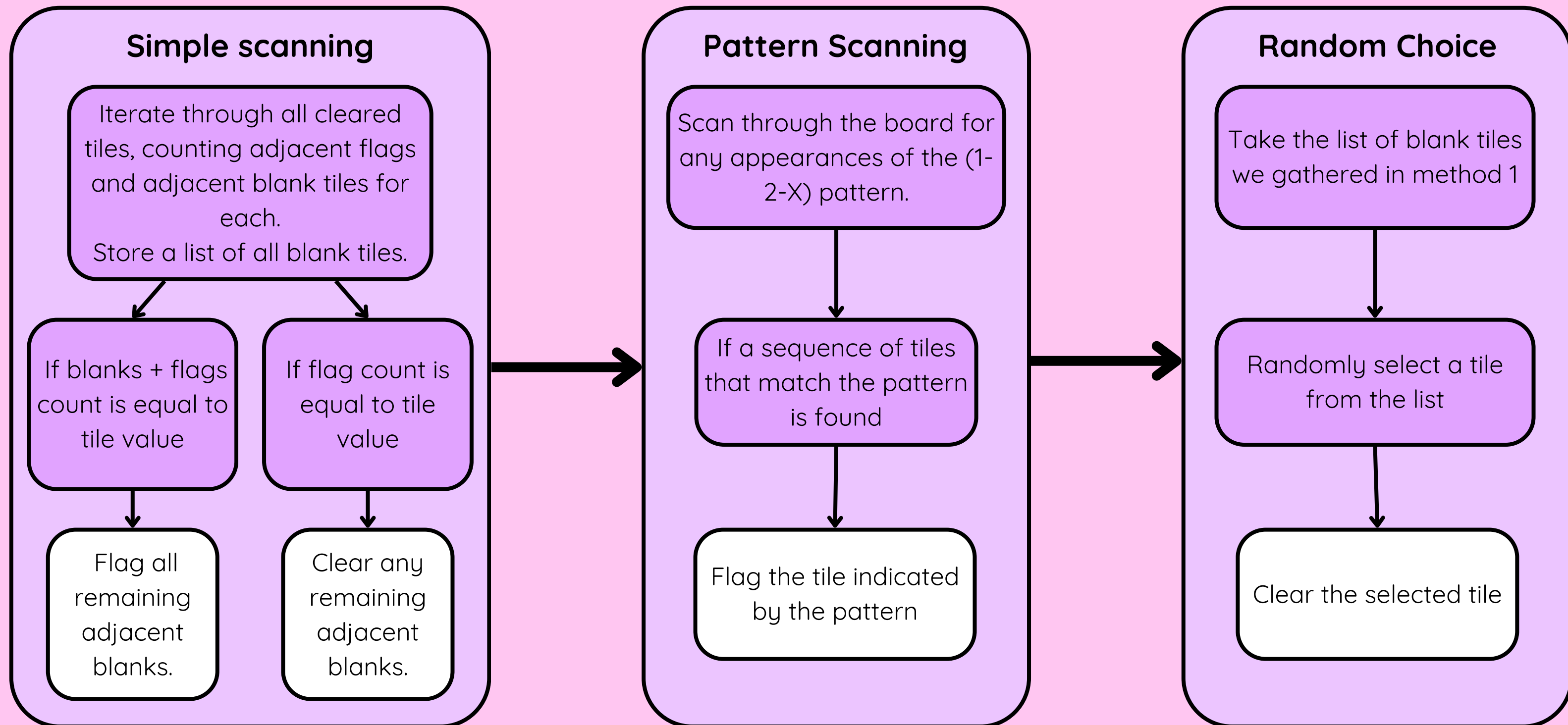
Pattern scanning has been implemented in this agent.

One of the possible ways to locate and flag mines is done using patterns. These are a specific series/configuration of tiles that have a set mine position whenever they appear. The pattern this agent uses is called the 1-2-X pattern, also known as the 1-2-1 pattern. The layout of this pattern is demonstrated in the image to the right.



Agent Design - Methods

There are 3 methods that the agent uses to decide its action. If one method doesn't have a viable action, then it moves to the next method



Problem Instances

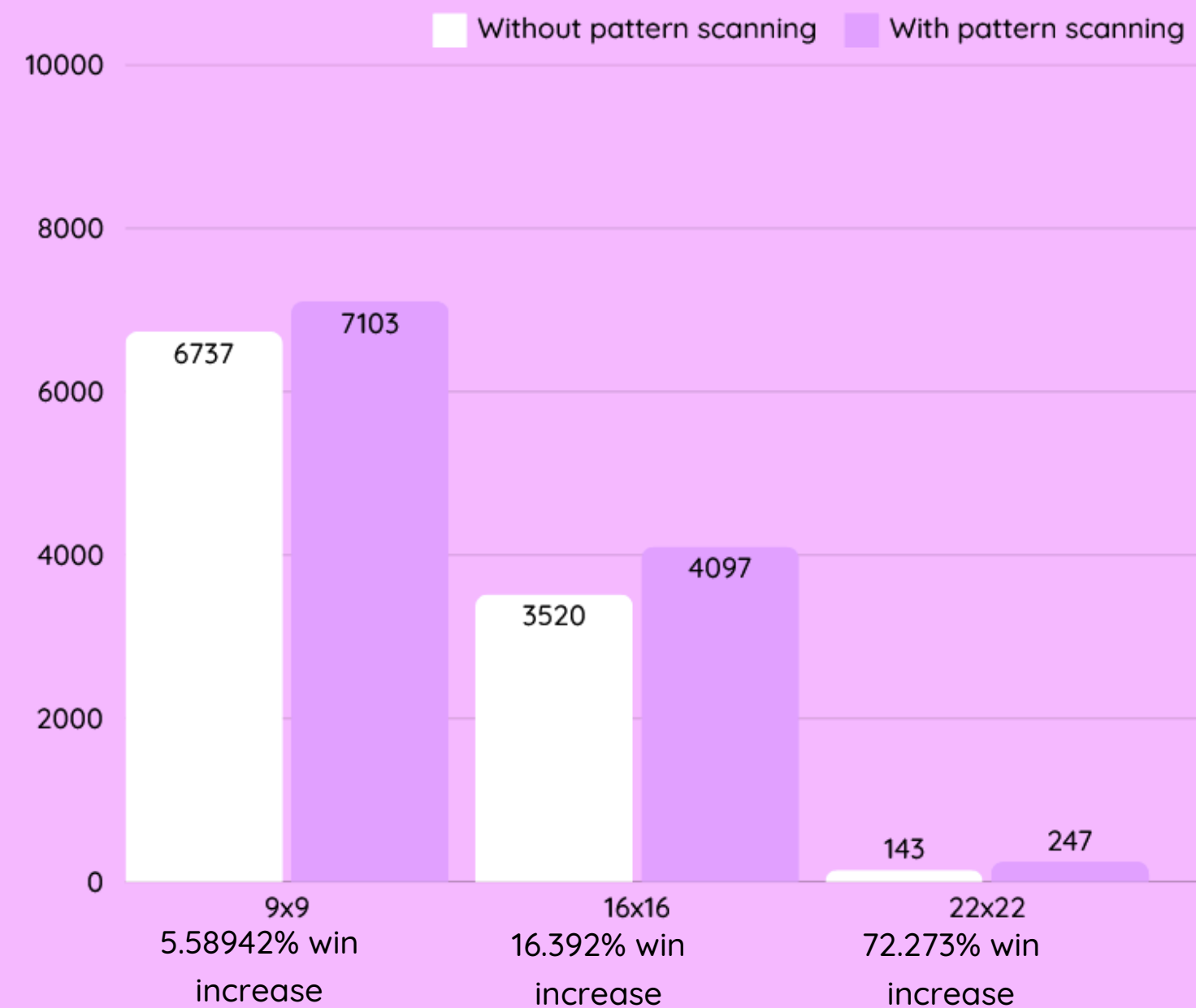
Modes

- Default
- No Guessing

Board sizes

- 9x9 board with 10 mines
- 16x16 board with 40 mines
- 22x22 board with 99 mines

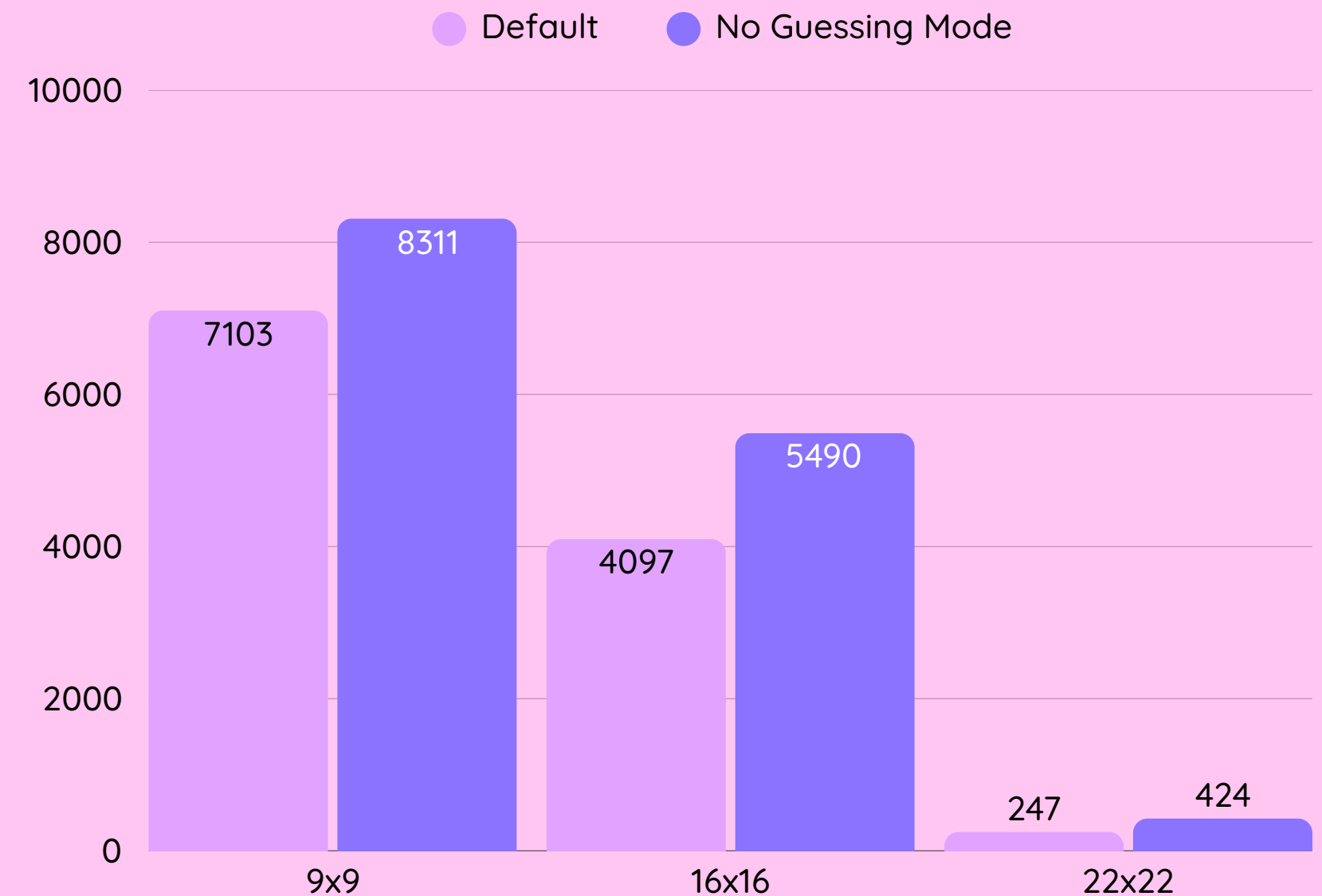
Results - The effects of 1-2-X pattern scanning



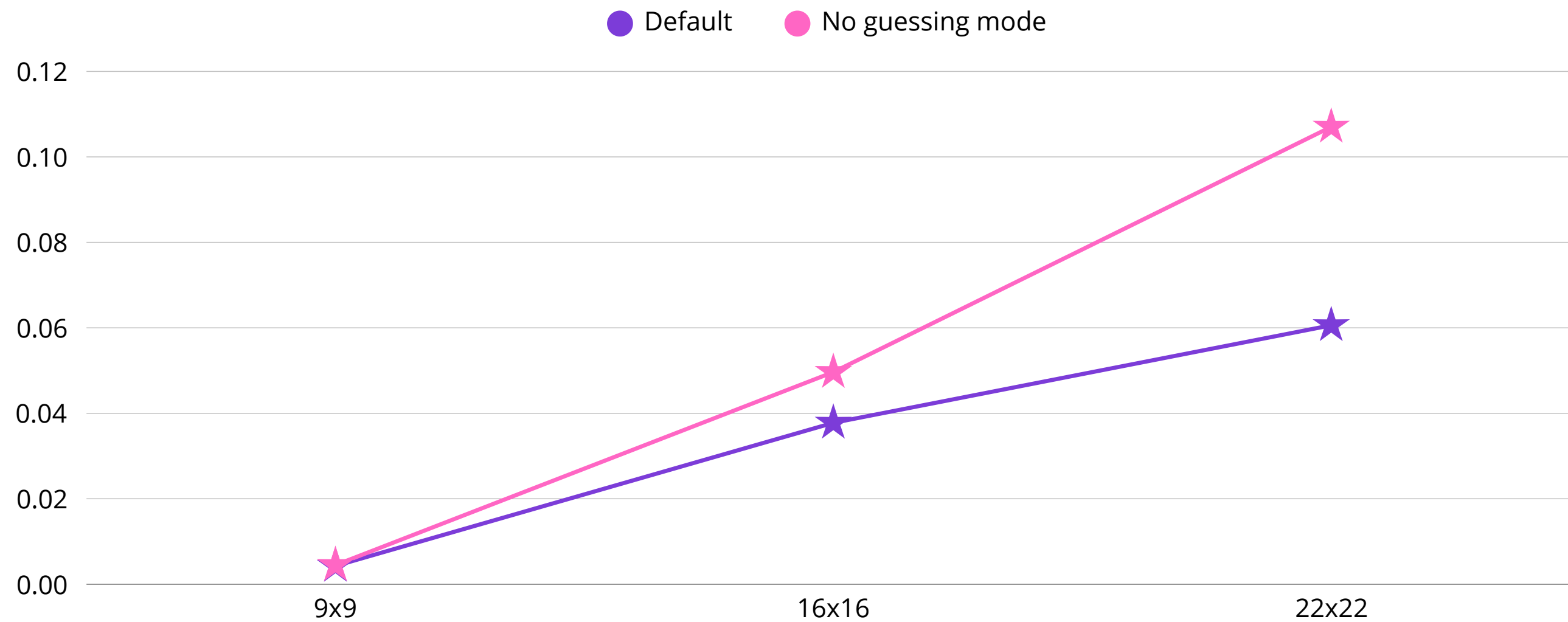
My first agent did not use pattern scanning and relied solely on scanning adjacent tiles or randomly choosing a tile. My second agent did implement pattern scanning. This chart shows the number of wins out of 10,000 runs for each agent. Below each set is the win percentage increase from not using pattern scanning to using pattern scanning

Results - The effects of ‘No Guessing’ Mode

Each board size setting was ran 10,000 time in both the default mode and the No guessing mode. This chart shows the difference in the number of wins when using each setting. As you can see, No guessing mode wins much more often than the default mode.



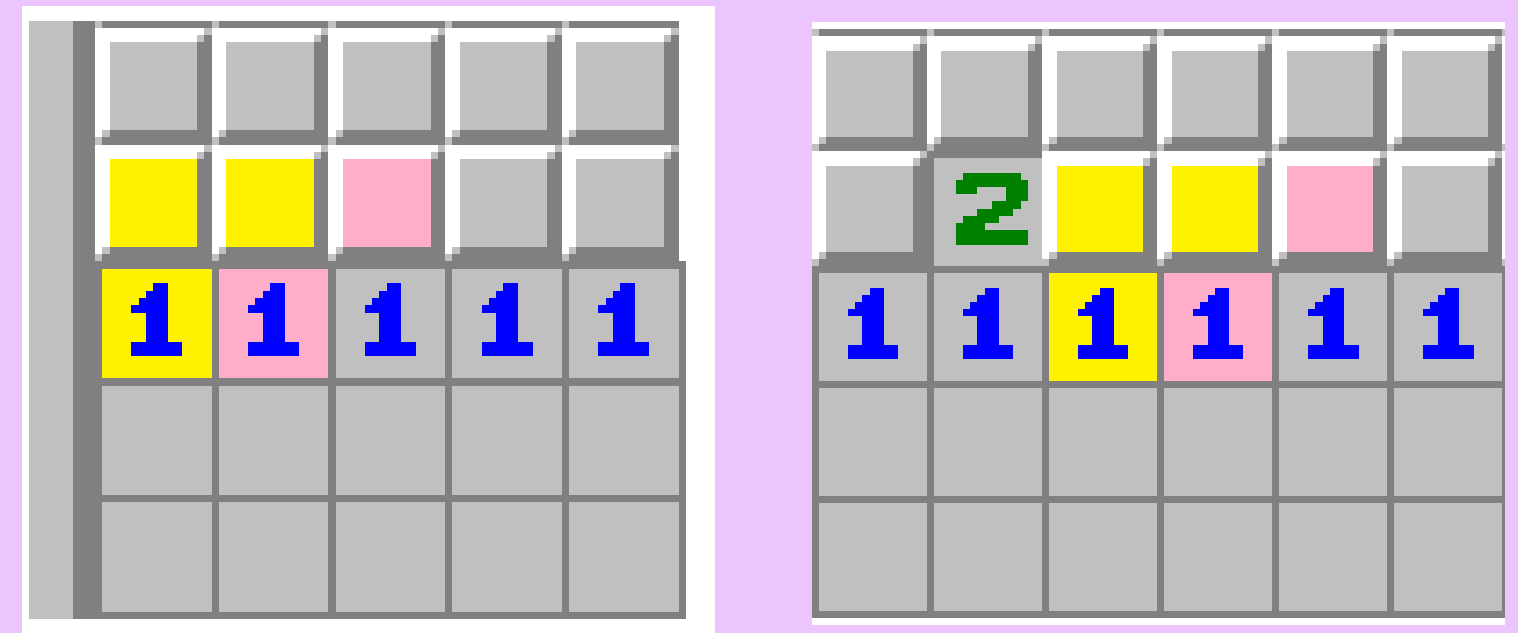
Results - Run time comparison



Possible Improvements

- Implement pattern scanning for other known sequences
- Alter code to shorten run time as much as possible
- Add an option to customize the board size and mine count

Possible pattern to add, referred to as the 1-1-X pattern



THANK YOU 🐣